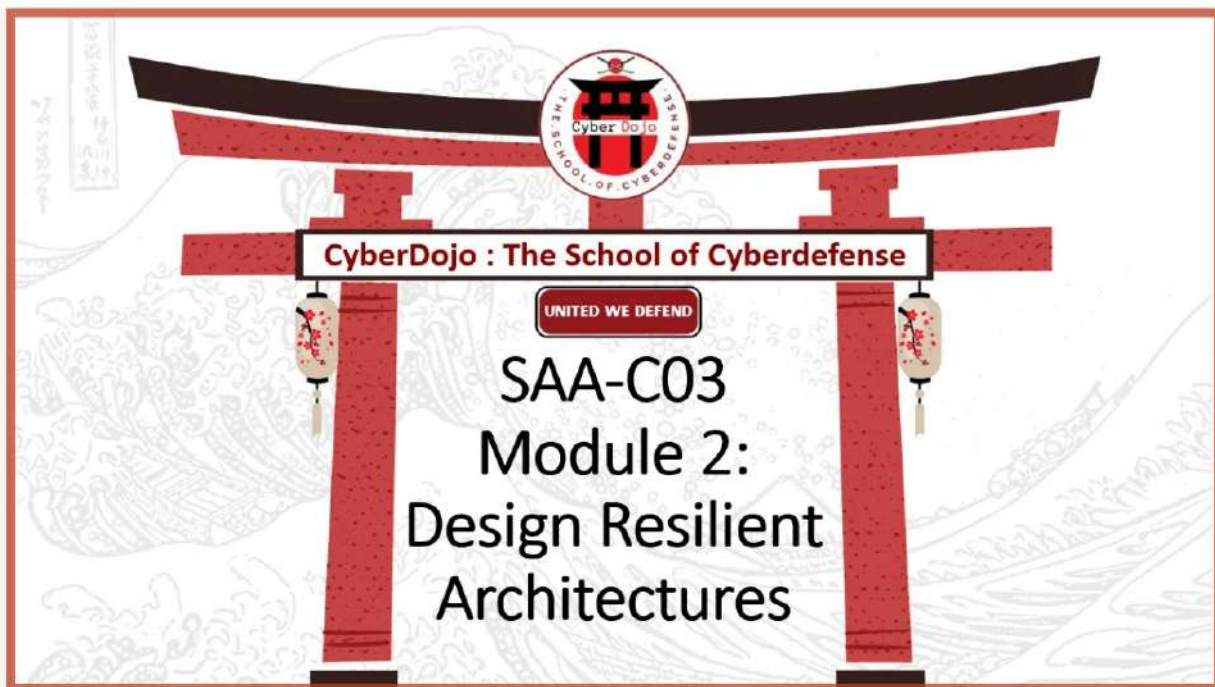
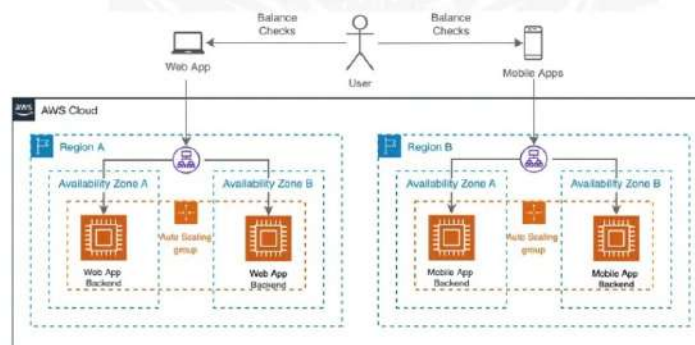




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Module 2: Design Resilient Architectures

- Section 2.1: Scalable & Loose-Coupled Architectures (SQS, SNS, EventBridge, Step Functions, Lambda)
- Section 2.2: Highly Available & Fault-Tolerant Architectures (Multi-AZ, Load Balancers, Route 53, DR Strategies, RDS HA)



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Securing Access to AWS Resources

- **Module 2:** Design Resilient Architectures (26% of exam)
- **Section 2.1:** Designing Scalable and Loose-Coupled Architectures
- **Topics Covered:**
 - Loose Coupling Concepts
 - Amazon SQS (Standard vs. FIFO)
 - Amazon SNS (Pub/Sub)
 - SNS + SQS Fan-out Pattern
 - Amazon EventBridge
 - AWS Step Functions
 - EC2 Scaling Strategies
 - Container Options (ECS, EKS, Fargate)
 - AWS Lambda



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Loose Coupling Concept - Why Decouple?

Definition: Designing systems where components have minimal dependencies on each other

Tightly Coupled (Monolithic) Problems:

Problem	Impact
Single point of failure	One component fails → entire system fails
Scaling together	Scale everything when only one component needs it
Deployment coordination	Update everything together
Technology lock-in	Hard to change one component

Loosely Coupled Benefits:

Benefit	How it helps
Fault isolation	Component failure doesn't cascade
Independent scaling	Scale only what needs scaling
Independent deployment	Update one service without others
Technology flexibility	Each component can use different tech

• **SAA Exam Focus:**

- Loose coupling = **Resilience + Scalability**
- Use queues, event buses, and async patterns
- Exam will present monolithic problems → decoupled solutions



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Amazon SQS - Simple Queue Service (Deep Dive)

Definition: Fully managed message queuing service for decoupling applications

Queue Types (MEMORIZE for SAA):

Feature	Standard Queue	FIFO Queue
Throughput	Unlimited (API calls/sec)	300 msg/sec (batch) / 3000 msg/sec (batch)
Ordering	Best-effort (not guaranteed)	Guaranteed (first-in-first-out)
Deduplication	No	Yes (content or ID based)
Exactly-once processing	At-least-once (possible duplicates)	Exactly-once
Use case	High throughput, order doesn't matter	Order matters (banking, ticketing)

Key SQS Concepts:

Concept	Description	Default
Visibility Timeout	Time message invisible after being read	30 seconds
Dead-Letter Queue (DLQ)	Queue for failed messages after max receives	Configurable
Message Retention	How long message stays in queue	4 days (1 min - 14 days)
Long Polling	Wait for messages (reduces empty responses)	Enabled
Message Size	Maximum message size	256 KB



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SQS Use Cases - Real World Patterns

Use Case 1: Decouple Web and App Tiers

User → Web Server → SQS Queue → App Server → Database

Benefit:

- Traffic spikes handled by queue (buffer)
- App servers process at their own pace
- If App servers fail, messages safe in queue

Use Case 2: Batch Processing

User Upload → API Gateway → SQS Queue → Lambda (batch)

Messages accumulate → Lambda processes in batches (e.g., 10 at a time)

Use Case 3: Buffering for Variable Workloads

Video Upload → SQS Queue → Transcoding Service

Queue absorbs variable upload rates
Transcoding service processes at steady rate

Common Exam Scenario:

- Problem: "Application experiences traffic spikes. Users get timeouts."
- Solution: Place SQS queue between web and app tiers (buffer)
- Wrong answer: "Increase EC2 instance size" (scaling, not decoupling)



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Amazon SNS - Simple Notification Service

Definition: Fully managed Pub/Sub messaging service

How SNS Works:

Publisher → SNS Topic → Subscribers

Subscriber types:

- SQS Queue
- Lambda Function
- Email (SNS)
- HTTP/HTTPS Endpoint
- Mobile Push

• **SAA Exam Focus:**

- SNS = **One to Many** (broadcast)
- SQS = **One to One** (point-to-point)
- Use SNS when multiple services need same message

SNS vs. SQS (Key Difference):

Feature	SQS	SNS
Model	Queue (point-to-point)	Pub/Sub (topic)
Consumers	One consumer pulls message	Multiple subscribers receive
Message persistence	Yes (retention)	No (delivered or lost)
Pull vs. Push	Pull (consumer polls)	Push (topic pushes)



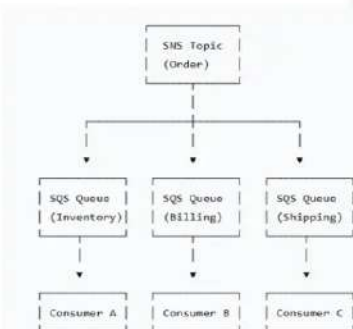
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SNS + SQS Fan-out Pattern (CRITICAL for SAA)

What is Fan-out? One SNS topic delivers same message to multiple SQS queues

Architecture:



Why This Pattern?

Benefit	Explanation
Decoupling	Each consumer processes independently
Fault tolerance	If one queue fails, others work
Different speeds	Each consumer can process at own pace
Replay capability	Messages persist in SQS

SAA Exam Focus:

- This is a **MUST KNOW** pattern for SAA
- SNS delivers to SQS → SQS provides durability + decoupling
- Each consumer gets its own queue



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Amazon EventBridge - Serverless Event Bus

Definition: Serverless event bus that connects applications using events from your own apps, AWS services, and SaaS partners

Event Sources:

Source Type	Examples
AWS Services	EC2 state change, S3 object created, CloudWatch events
Custom apps	Your own application events
SaaS partners	Datadog, Zendesk, Shopify, Salesforce

Real World:

- EC2 termination → EventBridge Rule → SNS Topic → Slack (via Lambda)

EventBridge vs. SNS vs. SQS:

Service	Best for
SQS	Simple queue, point-to-point, pull-based
SNS	Simple pub/sub, push to many
EventBridge	Complex routing, event filtering, SaaS integration

• **SAA Exam Focus:**

- EventBridge = **Event-driven architecture**
- Can filter events (e.g., only stopped EC2 instances)
- Built-in SaaS integrations (no custom code)



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AWS Step Functions - Serverless Orchestration

Definition: Serverless orchestration service to coordinate multiple AWS services into workflows

Error Handling Features:

Feature	What it does
Retry	Automatically retry failed steps
Catch	Handle specific errors (Go to error handler)
Choice	Branch based on condition (if/else)
Parallel	Run multiple branches simultaneously
Wait	Pause workflow for X seconds

Common Step Functions Pattern:

```
Start → Lambda (Validate Order)
      → DynamoDB (Save Order)
      → SQS (Send to Payment)
      → Lambda (Update Inventory)
      → SNS (Send Confirmation)
      → End
```

Error handling: Retry, Catch, Fallback to DLQ

• SAA Exam Focus:

- Step Functions = **Orchestration** (not individual tasks)
- Use for multi-step, long-running processes
- Visual workflow in AWS Console

• Example Use Cases:

- Order fulfillment (validation → payment → shipping)
- Video processing (upload → transcode → notify)
- ETL pipelines (extract → transform → load)



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EC2 Scaling Strategies

Vertical Scaling (Scale Up/Down):

Aspect	Details
What it is	Change instance size (t2.micro → t2.large)
When to use	Single instance applications, databases
Limitation	Maximum instance size (hard limit)
Downtime	Requires reboot/stop-start

Horizontal Scaling (Scale Out/In):

Aspect	Details
What it is	Add/remove more instances
When to use	Stateless applications, web tiers
Limitation	Must be stateless
Downtime	No downtime (add/remove behind load balancer)

• SAA Exam Focus:

Question: "Web application experiences variable traffic"

✓ CORRECT: Horizontal scaling (add more instances)

✗ WRONG: Vertical scaling (limited by largest instance)

Question: "RDS database needs more performance"

✓ CORRECT: Vertical scaling (scale up instance type)

⚠ POSSIBLE: Read Replicas (horizontal for reads)



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EC2 Auto Scaling - Deep Dive

Auto Scaling Components:

Component	Purpose
Launch Template	AMI, instance type, SG, user data
Auto Scaling Group (ASG)	Min/Max/Desired capacity, VPC/subnets
Scaling Policy	When to scale (target tracking, simple, scheduled)

SAA Exam Focus:

- ASG = **Health checks** (replace unhealthy instances)
- ASG = **Fleet management** (maintain desired count)
- Cooldown period = Wait time between scaling actions

Scaling Policy Types:

Type	How it works	Best for
Target Tracking	Maintain metric at target (e.g., CPU = 50%)	Simplicity, dynamic workloads
Simple Scaling	Add/remove fixed number when alarm triggers	Predictable patterns
Step Scaling	Add/remove based on alarm severity	Granular control
Scheduled Scaling	Scale at specific times	Predictable patterns (lunch rush)



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Container Options - ECS, EKS, Fargate

Container Orchestration Comparison:

Service	Best for	Management
ECS	AWS-native, simple container workloads	Managed control plane
EKS	Kubernetes expertise, multi-cloud	Managed control plane
Fargate	Serverless containers (no EC2)	Fully managed (serverless)

• SAA Exam Focus:

- Fargate = **Serverless containers** (no EC2 management)
- Use ECS when you're AWS-focused
- Use EKS when you need Kubernetes (portability)

ECS vs. EKS Decision:

Factor	Choose ECS	Choose EKS
Kubernetes experience	No	Yes
AWS integration	Deep (IAM, CloudWatch)	Standard
Multi-cloud	No (AWS only)	Yes
Complexity	Lower	Higher

Launch Types:

Launch Type	Who manages EC2	Use Case
Fargate	AWS (serverless)	No infrastructure management
EC2	You	Need control, spot instances, larger workloads



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AWS Lambda - Event-Driven Compute

Definition: Serverless, event-driven compute service (no servers to manage)

Lambda Characteristics (MEMORIZE):

Characteristic	Details
Max execution time	15 minutes (900 seconds)
Max memory	10 GB (10,240 MB)
Ephemeral storage	512 MB to 10 GB
Deployment package	50 MB (zip), 250 MB (container)
Concurrency limit	1000 per region (soft limit)

Common Lambda Triggers (SAA Favorites):

Trigger	Use Case
S3	Image processing on upload
API Gateway	Serverless backend
DynamoDB Streams	React to database changes
SQS	Process queue messages
EventBridge	Scheduled jobs (cron)
CloudWatch Logs	Log processing



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AWS Lambda - Event-Driven Compute

Lambda vs. EC2 Decision:

Factor	Lambda	EC2
Execution time	<15 minutes	Unlimited
State	Stateless (use external DB)	Can be stateful
Cold start	Possible (milliseconds to seconds)	Always running
Cost	Pay per invocation	Pay per hour
Best for	Event-driven, intermittent	Long-running, steady

• SAA Exam Focus:

- Lambda = **Event-driven, short-running, stateless**
- Use Lambda for intermittent workloads (cost-effective)
- Use EC2 for long-running, predictable workloads



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When to Use What - Compute Decision Tree

START: What is your workload?

```
├ Long-running (>15 min) or predictable steady-state?
├   └ YES → EC2 (or EC2 + Auto Scaling)
├ Containerized application?
├   └ YES → Do you need Kubernetes?
├       └ YES → EKS
├       └ NO → ECS
├ Want serverless (no EC2 management)?
├   └ YES → Fargate (containers) or Lambda (Functions)
├ Event-driven, intermittent, short (<15 min)?
├   └ YES → Lambda (cheapest for low volume)
```

Decision Matrix:

Workload Type	Best Service	Why
Web server (steady traffic)	EC2 + ASG	Predictable, long-running
Web server (spiky)	Lambda + API Gateway	Pay per request
Microservices (containers)	ECS + Fargate	Serverless containers
Batch processing (nightly)	Lambda (if <15 min)	No idle cost
Batch processing (hours)	EC2 Spot	Cheapest compute
ML training	EC2 (GPU instances)	Specialized hardware



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Lab/Hands-on Guide - SQS, Lambda, SNS

Lab 1: SQS Queue + Lambda Trigger

1. Create SQS queue (Standard)
2. Create Lambda function (print event)
3. Add SQS trigger to Lambda
4. Send test message to SQS
5. Verify Lambda processed message

Lab 2: SNS Topic + Multiple Subscribers

1. Create SNS topic
2. Create two SQS queues
3. Subscribe both queues to SNS topic
4. Publish message to topic
5. Check both queues received message

Lab 3: EventBridge Scheduled Lambda

1. Create Lambda function
2. Create EventBridge rule (cron schedule)
3. Target: Lambda function
4. Test (wait for schedule or run manually)

Lab 4: Step Functions Workflow

1. Open Step Functions console
2. Create state machine (visual editor)
3. Add Lambda, Wait, Choice states
4. Execute workflow
5. View execution history



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Section 2.1 Summary - Scalable & Loose-Coupled

Service	Purpose	Key Exam Point
SQS Standard	High-throughput queue	At-least-once, unordered
SQS FIFO	Ordered queue	Exactly-once, limited throughput
SNS	Pub/Sub broadcast	Push to many subscribers
SNS + SQS	Fan-out pattern	Durable, decoupled broadcast
EventBridge	Event bus	SaaS integration, filtering
Step Functions	Workflow orchestration	Long-running, error handling
EC2 Auto Scaling	Horizontal scaling	Target tracking (simplest)
Fargate	Serverless containers	No EC2 management
Lambda	Event-driven compute	<15 min, stateless



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Section 2.2: Designing Highly Available and Fault-Tolerant Architectures

- **Module 2:** Design Resilient Architectures (26% of exam)
- **Section 2.2:** Designing Highly Available and Fault-Tolerant Architectures
- **Topics Covered:**
 - AWS Global Infrastructure (Regions, AZs, Edge Locations)
 - Multi-AZ Strategy
 - Load Balancer Types (ALB, NLB, GWLB)
 - Route 53 Routing Policies
 - Disaster Recovery Strategies (RPO/RTO)
 - DR Patterns: Backup & Restore, Pilot Light, Warm Standby, Multi-Site
 - RDS High Availability (Multi-AZ vs. Read Replicas)
 - Aurora High Availability
 - DynamoDB Global Tables
 - S3 Cross-Region Replication (CRR)
 - EBS Snapshots for DR



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AWS Global Infrastructure - Regions, AZs, Edge Locations

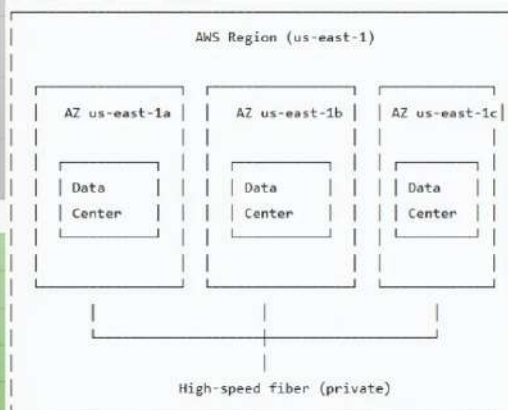
Three Components (MEMORIZE):

Component	Description	How Many?
Region	Geographic area (e.g., us-east-1)	30+
Availability Zone (AZ)	Isolated data center(s) within region	90+
Edge Location	CDN cache location (CloudFront)	400+

Selection Factors:

Factor	Why it matters
Compliance	Data residency laws (GDPR)
Latency	Proximity to users
Feature availability	New services roll out gradually
Pricing	Costs vary by region

AZ Architecture:



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Multi-AZ Strategy - The Foundation of HA

What is Multi-AZ? Deploying resources across two or more Availability Zones

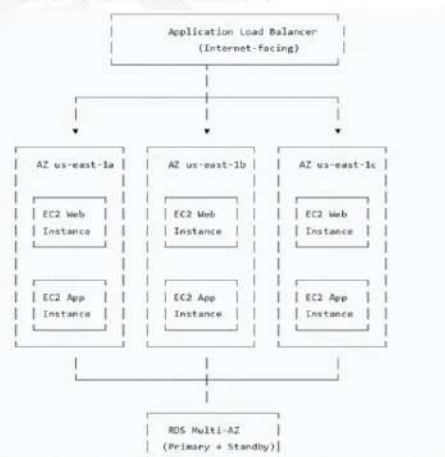
Why Multi-AZ?

Without Multi-AZ	With Multi-AZ
Single AZ failure = application down	AZ failure = other AZ takes over
No redundancy	Built-in failover
Lower availability (99.9% vs 99.99%)	Higher SLA

SAA Exam Focus:

- **Minimum 2 AZs** for production workloads
- **3 AZs** for higher availability
- ALB + Auto Scaling across AZs = **Resilient web tier**
- RDS Multi-AZ = **Resilient database tier**

Multi-AZ Architecture Pattern:



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Load Balancer Types - ALB, NLB, GWLB

Three Types (MEMORIZE for SAA):

Type	Layer	Protocols	Best for
Application Load Balancer (ALB)	7 (HTTP/HTTPS)	HTTP, HTTPS, gRPC	Web apps, microservices, path-based routing
Network Load Balancer (NLB)	4 (TCP/UDP)	TCP, UDP, TLS	Ultra-low latency, millions of requests, static IP
Gateway Load Balancer (GWLB)	3/4	IP	Deploying virtual appliances (firewalls, IDS)

ALB Features (Most common in SAA):

Feature	What it does
Path-based routing	/api/* → API service, /images/* → Image service
Host-based routing	api.example.com → API service, www.example.com → Web service
Sticky sessions	Route same user to same instance
Authentication	Integrate with Cognito, OIDC
WebSocket	Support real-time apps



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Load Balancer Types - ALB, NLB, GWLB

NLB Features:

Feature	What it does
Static IP	Fixed IP address (ALB doesn't have)
Elastic IP	Assign specific EIP
Ultra-low latency	~100 microseconds (vs ALB milliseconds)
TLS termination	Offload SSL at load balancer

• SAA Exam Focus:

- ALB = HTTP/HTTPS traffic (web apps)
- NLB = TCP/UDP + need static IP
- GWLB = Virtual appliances (firewalls)



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ALB Deep Dive - Listeners, Target Groups, Health Checks

ALB Components:

```

Internet → ALB (Listener:80) → Target Group → EC2 Instances
↓
Listener Rule: /api/* → Target Group A (API servers)
Listener Rule: /images/* → Target Group B (Image servers)
Listener Rule: default → Target Group C (Web servers)
    
```

Target Types:

Target Type	Example	Use Case
Instance	i-1234567890	EC2 instances (by ID)
IP	10.0.1.5	On-premises or peered VPC
Lambda	Function ARN	Serverless backend

Health Check Configuration:

Setting	Description	Default
Health check path	Endpoint to check (e.g., /health)	/
Healthy threshold	Consecutive successes to mark healthy	3
Unhealthy threshold	Consecutive failures to mark unhealthy	3
Interval	How often to check	30 seconds
Timeout	Max wait for response	5 seconds



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NLB Use Cases - Static IP, TCP/UDP, Ultra-low latency

When to choose NLB over ALB:

Requirement	Why NLB
Static IP address	ALB IP changes; NLB has static IP + can use Elastic IP
TCP/UDP traffic	Gaming, VoIP, IoT protocols
Ultra-low latency	NLB ~100 microseconds vs ALB milliseconds
Millions of requests/sec	NLB scales to handle massive throughput
TLS termination at scale	NLB can terminate millions of TLS connections

NLB vs. ALB Decision:

Factor	ALB	NLB
HTTP/HTTPS	✓ Yes	✓ Yes (but basic)
WebSocket	✓ Yes	✗ No
Path-based routing	✓ Yes	✗ No
Static IP	✗ No	✓ Yes
TCP/UDP	✗ No	✓ Yes
Cost	Higher (per rule)	Lower

NLB Architecture:

Internet → NLB (Static IP: 203.0.113.10) → Target Group → EC2 Instances
 ↓
 Can preserve client IP (via Proxy Protocol)



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Route 53 Routing Policies (CRITICAL for SAA)

Routing Policies (MEMORIZE each use case):

Policy	How it works	Use Case
Simple	One record, one destination	Single resource (no HA)
Weighted	Split traffic by percentage	A/B testing, blue/green
Latency	Route to lowest latency region	Global users, performance
Failover	Primary (active) → Secondary (passive)	Disaster recovery
Geolocation	Route based on user location	Localized content, compliance
Geoproximity	Route based on geographic distance	Shift traffic between regions
Multi-value	Return multiple healthy IPs	Simple load balancing (no ELB)



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Route 53 Routing Policies (CRITICAL for SAA)

Weighted Routing Example:

New version: 10% of traffic
 Old version: 90% of traffic
 Gradually shift: 10% → 25% → 50% → 75% → 100%

Failover Routing (Active-Passive):

Primary Region (us-east-1): Healthy → All traffic here
 Secondary Region (us-west-2): Standby → Traffic only if primary fails
 Health check on primary → Failover if unhealthy

SAA Exam Focus:

- **Weighted** = Gradual rollout, testing
- **Latency** = Performance optimization
- **Failover** = Disaster recovery
- **Geolocation** = Localized content (GDPR compliance)

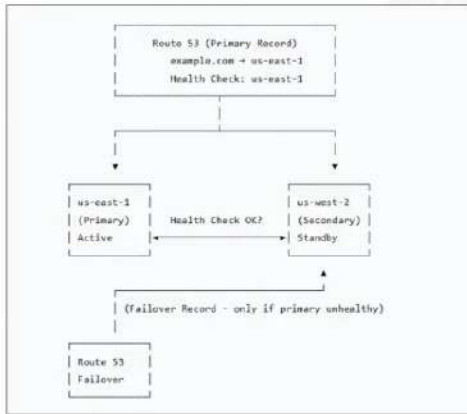


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Route 53 Failover - Active-Passive and Active-Active

Active-Passive (Failover):



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Active-Active (Weighted with Health Checks):

- example.com → Weighted Record: us-east-1 (50%) + us-west-2 (50%)
- Both with health checks → If one fails, traffic shifts to healthy region

SAA Exam Focus:

- **Active-Passive** = Lower cost (secondary region idle)
- **Active-Active** = Higher cost but better performance (both regions active)
- Always configure **health checks** for failover to work

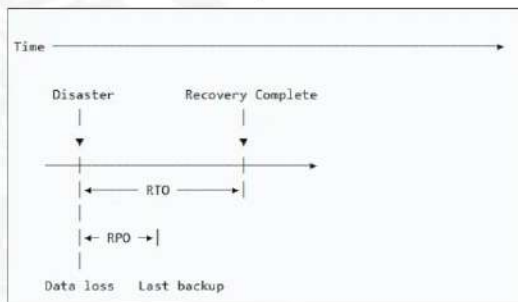
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Disaster Recovery Concepts - RPO and RTO

Definitions (MEMORIZE - heavily tested):

Term	Definition	Example
RPO (Recovery Point Objective)	Maximum acceptable data loss (time between last backup and disaster)	"Lose up to 5 minutes of data"
RTO (Recovery Time Objective)	Maximum acceptable downtime (time to restore service)	"Back up in 4 hours"
MTD (Maximum Tolerable Downtime)	Total downtime business can survive	"Business fails after 24 hours"

RPO and RTO Relationship:



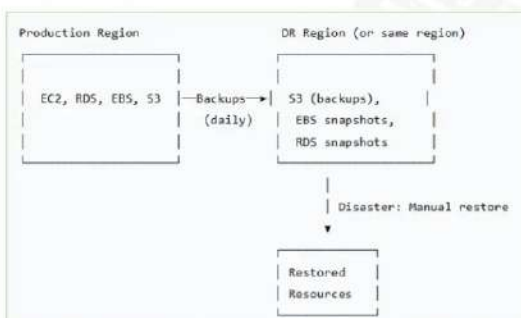
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DR Strategy 1 - Backup & Restore (Highest RPO/RTO)

Description: Regular backups; restore from backups after disaster

Architecture:



Characteristics:

Aspect	Value
RPO	Hours to days (depends on backup frequency)
RTO	Hours to days (manual restore time)
Cost	Lowest (\$)
Complexity	Lowest
Automation	Minimal (manual failover)

When to use:

- Non-critical applications
- Low RPO/RTO requirements (can tolerate hours of loss/downtime)
- Budget-constrained



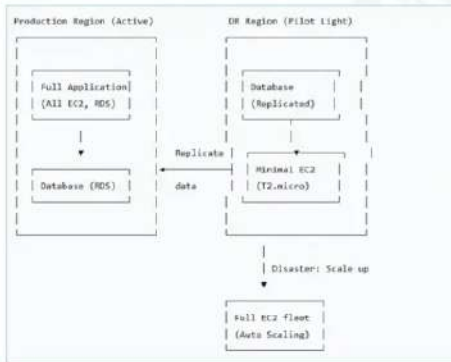
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DR Strategy 2 - Pilot Light (Better RPO/RTO)

Description: Minimal core services running in DR region; scale up on disaster

Architecture:



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Characteristics:

Aspect	Value
RPO	Minutes (database replication)
RTO	10-30 minutes (scale up time)
Cost	Low to Medium (\$\$)
Complexity	Medium
Automation	Partial (manual scaling trigger)

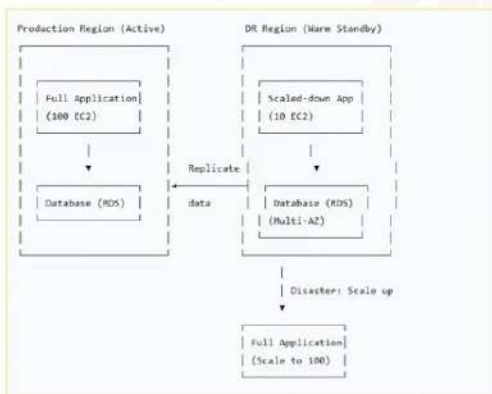
• **When to use:**

- Can tolerate ~30 minutes downtime
- Need better RPO than backup/restore
- Moderate budget

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DR Strategy 3 - Warm Standby (Low RPO/RTO)

Description: Scaled-down but fully functional environment in DR region



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Characteristics:

Aspect	Value
RPO	Seconds (synchronous or near-sync replication)
RTO	5-15 minutes (scale up time)
Cost	Medium to High (\$\$\$)
Complexity	Medium to High
Automation	Partial (scaling may be automated)

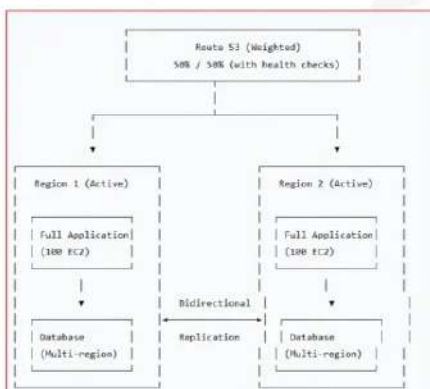
• **When to use:**

- Business-critical applications
- Need RTO under 30 minutes
- Can afford running scaled-down environment

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DR Strategy 4 - Multi-Site Active-Active

Description: Traffic distributed across multiple regions; all regions active



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Characteristics:

Aspect	Value
RPO	0 seconds (no data loss)
RTO	Seconds to minutes (automatic failover)
Cost	Highest (\$\$\$\$)
Complexity	High
Automation	Full (automatic failover)

• **When to use:**

- Mission-critical applications
- Zero downtime requirement
- Unlimited budget (or business justifies cost)

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DR Strategies Comparison Table (MEMORIZE)

Strategy	RPO	RTO	Cost	Complexity	When to Use
Backup & Restore	Hours-Days	Hours-Days	\$	Low	Non-critical, low budget
Pilot Light	Minutes	10-30 min	\$\$	Medium	Can tolerate ~30 min downtime
Warm Standby	Seconds	5-15 min	\$\$\$	Medium-High	Business-critical
Multi-Site Active-Active	0 seconds	Seconds	\$\$\$\$	High	Mission-critical, zero downtime



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RDS High Availability - Multi-AZ vs. Read Replicas

Multi-AZ (High Availability - NOT performance):

Aspect	Details
Purpose	Fallover (HA), not read scaling
Replication	Synchronous
Standby	One standby in different AZ
Fallover	Automatic (30-60 seconds)
Use case	Production databases needing HA

• SAA Exam Focus:

- **Multi-AZ** = Answer when question asks for "High Availability" or "Automatic failover"
- **Read Replicas** = Answer when question asks for "Read-heavy workload" or "Offload reads"

Read Replicas (Read Scaling - NOT HA):

Aspect	Details
Purpose	Offload read traffic
Replication	Asynchronous
Number	Up to 5 per primary
Fallover	Manual (promote to primary)
Use case	Read-heavy workloads

Can use BOTH together!

```

Primary (us-east-1a) ←sync→ Standby (us-east-1b) [Multi-AZ]
|
├─async→ Read Replica 1
└─async→ Read Replica 2 (cross-region)
    
```



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RDS High Availability - Multi-AZ vs. Read Replicas

Comparison Table (MEMORIZE):

Feature	Multi-AZ	Read Replicas
Primary purpose	High Availability	Read scalability
Replication type	Synchronous	Asynchronous
Standby accepts traffic?	No (standby only)	Yes (read traffic)
Automatic failover?	Yes	No (manual promotion)
Can be in different region?	No (same region)	Yes (cross-region)
Cost	2x (primary + standby)	1x + replica cost



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Aurora High Availability - Aurora Replicas & Global Database

Aurora Replicas (Up to 15):

Feature	Details
Purpose	Read scaling + HA (failover target)
Replication	Asynchronous (milliseconds)
Failover	Automatic (30 seconds)
Auto-scaling	Storage auto-scales (10GB to 128TB)
Durability	6 copies across 3 AZs

Aurora Global Database:

Feature	Details
Purpose	Cross-region DR
Replication	<1 second (typical)
RPO	1 second
RTO	1 minute
Regions	1 primary + up to 5 secondary

• SAA Exam Focus:

- Aurora = **Better than RDS** (performance + HA)
- Aurora Global Database = **Cross-region DR** (1 sec RPO, 1 min RTO)
- 15 Aurora Replicas (vs. 5 for RDS)



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DynamoDB High Availability - Global Tables

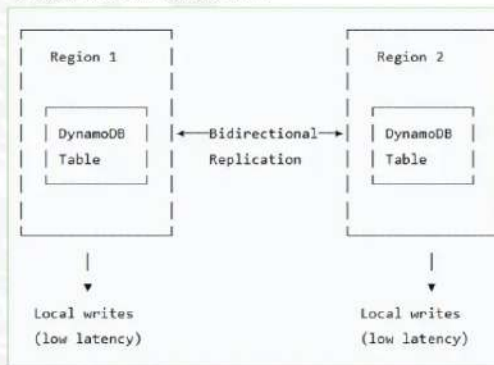
DynamoDB Built-in HA:

- Data automatically replicated across 3 AZs (within region)
- 99.999% availability SLA (5 nines)

DynamoDB Global Tables:

Feature	Details
Purpose	Multi-region, active-active replication
Replication	Fully managed, automatic
RPO	0 seconds (no data loss)
RTO	Seconds (automatic)
Use case	Global applications, DR

Global Tables Architecture:



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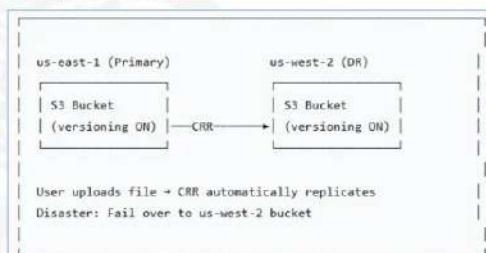
S3 Cross-Region Replication (CRR)

What is CRR? Automatically replicate objects from one S3 bucket to another bucket in different region

CRR Requirements:

Requirement	Details
Source bucket	Must enable versioning
Destination bucket	Must enable versioning
IAM role	Permission to replicate
Use cases	DR, compliance, low-latency access

CRR Architecture:



• SAA Exam Focus:

- CRR = **Asynchronous replication** (not real-time, but fast)
- Use for **DR** (replicate to another region)
- Use for **Compliance** (store copies in multiple regions)
- Use **S3 Replication Time Control (RTC)** for predictable replication (15 min SLA)



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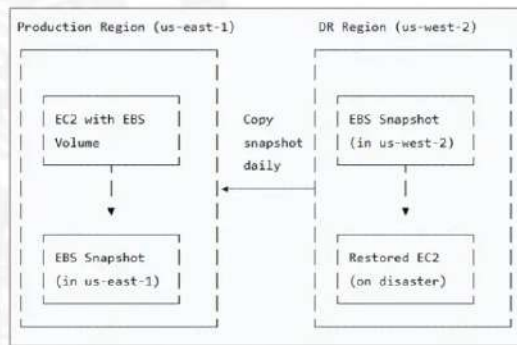
EBS Snapshots for Disaster Recovery

What are EBS Snapshots? Incremental backups of EBS volumes stored in S3

Snapshot Features:

Feature	Benefit
Incremental	Only changed blocks (cost-effective)
Cross-region copy	Copy snapshot to DR region
Fast snapshot restore	No warm-up time (costs extra)
Snapshot Archive	Cheaper for long-term retention (90+ days)
Recycle Bin	Recover accidentally deleted snapshots

EBS Snapshot DR Pattern:



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Service Quotas & Throttling - Design Considerations

Service Quotas (Limits):

Service	Default Quota	Can Increase?
EC2 instances per region	5 (t2.micro) to 384 (m5.8xlarge)	✓ Yes (request)
EBS volume size	16 TB (gp3, io1)	✓ Yes
S3 buckets per account	100	✓ Yes
Lambda concurrency	1000	✓ Yes
RDS instances per region	40	✓ Yes

Throttling Handling Strategies:

Strategy	How it works	When to use
Exponential Backoff	Retry with increasing delays	Standard approach (built into AWS SDKs)
Jitter	Add randomness to retry timing	Avoid "thundering herd"
Request batching	Group multiple requests	API operations (e.g., DynamoDB BatchWriteItem)
Queue buffering	SQS between components	Variable workload patterns



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Workload Visibility - AWS X-Ray

What is X-Ray? Service to trace and analyze requests as they travel through distributed applications

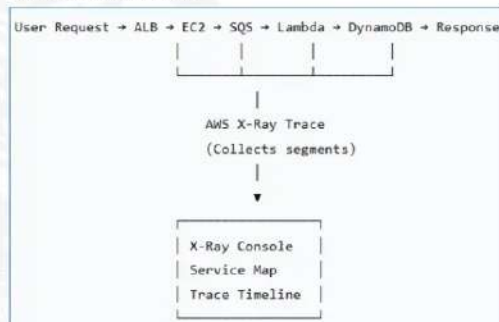
X-Ray Benefits:

Benefit	Description
End-to-end tracing	Follow request through multiple services
Performance analysis	Find bottlenecks
Error detection	Identify failing components
Service map	Visualize application architecture

• **SAA Exam Focus:**

- X-Ray = **Distributed tracing**
- Use to debug microservices, serverless applications
- Identify **latency bottlenecks**
- No code changes needed (integrate with SDK)

X-Ray Architecture:



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Lab/Hands-on Guide - High Availability & DR

Lab 1: Multi-AZ Web Application

1. Create VPC with 2 public subnets (different AZs)
2. Create Application Load Balancer (Internet-facing)
3. Create Auto Scaling Group across both AZs
4. Launch EC2 instances (web servers)
5. Test: Stop one AZ instance → Traffic still flows

Lab 2: RDS Multi-AZ vs. Read Replicas

1. Launch RDS MySQL with Multi-AZ enabled
2. Create Read Replica in different AZ
3. Simulate failover (reboot with failover)
4. Connect to Read Replica for queries

Lab 3: Route 53 Failover Routing

1. Create two EC2 instances (different regions)
2. Create Route 53 records: Primary + Secondary
3. Configure health checks
4. Stop primary instance → Verify failover

Lab 4: EBS Snapshots for DR

1. Create EC2 instance with EBS volume
2. Create snapshot of volume
3. Copy snapshot to another region
4. Restore volume from snapshot in DR region



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Section 2.2 Summary - High Availability & Fault Tolerance

Concept	Key Takeaway
Multi-AZ	Deploy across ≥2 AZs for HA
ALB vs. NLB	ALB = HTTP, NLB = TCP/UDP + static IP
Route 53	Weighted (testing), Failover (DR), Latency (performance)
RPO vs. RTO	RPO = data loss, RTO = downtime
DR Strategies	Backup → Pilot Light → Warm Standby → Multi-Site (cost ↑, RPO/RTO ↓)
RDS Multi-AZ	Synchronous, automatic failover (HA, not read scaling)
RDS Read Replicas	Asynchronous, manual promotion (read scaling)
Aurora Global DB	Cross-region DR, 1 sec RPO
DynamoDB Global Tables	Active-Active cross-region
S3 CRR	Cross-region replication for DR
EBS Snapshots	Incremental backups, copy to DR region
X-Ray	Distributed tracing for microservices



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